

Network Threats & Mitigation & Physical Security Risk

Task 1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Common Network Threats

Threat	Description	Example
Phishing	Fake emails, SMS, or websites used to steal login credentials.	A fake Paytm SMS asking users to verify their account by clicking a fraudulent link.
DDoS (Distributed Denial of Service)	Flooding a server with massive traffic to make it unavailable.	An Instagram outage caused by heavy malicious traffic preventing users from accessing the service.
Man-in-the-Middle (MITM)	An attacker intercepts communication between two parties.	Connecting to an unsecured public Wi-Fi while using WhatsApp or banking apps, allowing traffic interception if protections are absent.
Malware	Malicious software that damages or steals information.	Downloading a fake APK claiming to be WhatsApp that installs malware on an Android phone.
Password Attack	Attempting to guess or steal passwords using various methods.	Attackers trying many password combinations to access an Instagram account.

Task 2.

Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

Constraint: The script should print the number of attempts it took to guess the correct PIN.

```
correct_pin = "4827"
attempts = 0

for i in range(10000):
    guess = str(i).zfill(4)
    attempts += 1

    if guess == correct_pin:
        print("PIN Found:", guess)
        print("Attempts:", attempts)
        break
```

The program checks every PIN from **0000** to **9999**.

It compares each guess with the preset PIN.

When the correct PIN is found, it prints the PIN and the number of attempts.

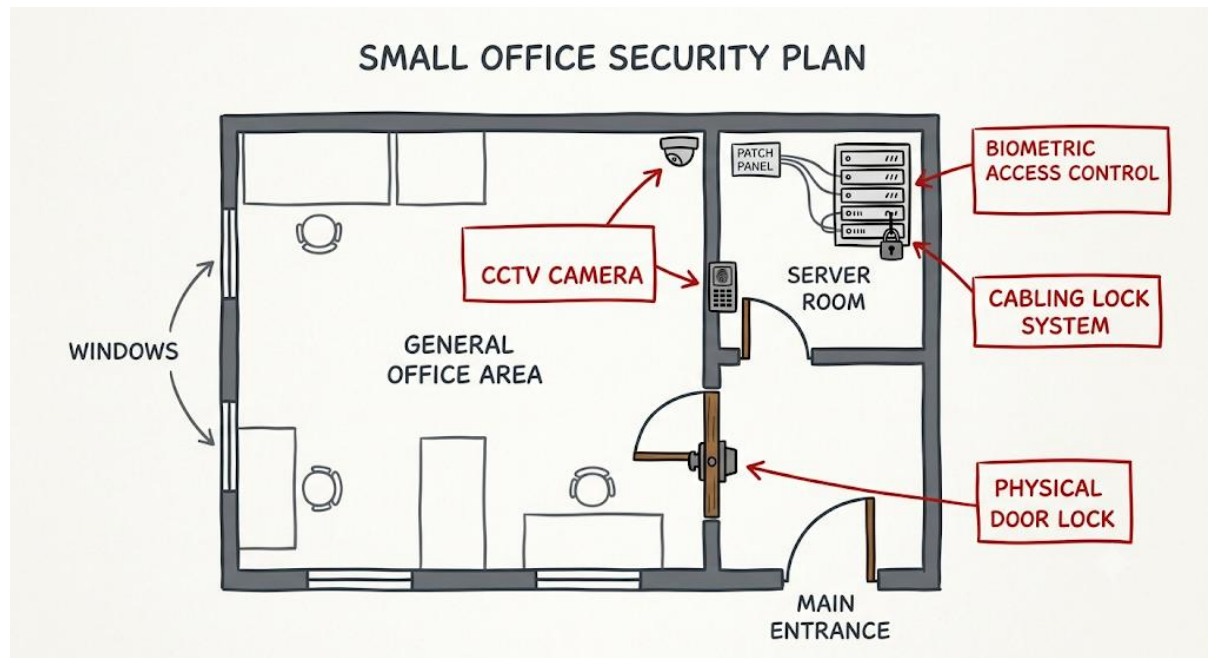
Task 3.

List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Risk	Mitigation
Unauthorized access	Install biometric or RFID access control.
Fire	Install smoke detectors and automatic fire suppression systems.
Power failure	Use UPS systems and backup generators.
Theft or equipment tampering	Install CCTV cameras and keep servers in locked racks.

Task 4.

Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



Task 5.

Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

Security Checklist for a College Computer Lab

1. Use strong passwords and enable multi-factor authentication (MFA) for administrator accounts.
2. Keep Windows, Linux, network devices, and antivirus software updated with the latest security patches.
3. Configure firewalls and install antivirus software on all lab computers.
4. Regularly back up important student data and network configurations.
5. Restrict access to the server room using biometric or RFID entry and monitor it with CCTV.

6. Lock networking equipment, switches, routers, and cables to prevent theft or tampering.
7. Train students and staff to recognize phishing emails, suspicious websites, and unsafe USB devices, and report security incidents immediately.